

Executive Summary

This report outlines a comprehensive data science approach to diagnosing, analyzing, and predicting supply chain inefficiencies, specifically focusing on late delivery risks and their financial impact. By transitioning from descriptive analytics to predictive machine learning, this project aims to empower business stakeholders to proactively mitigate delivery delays, optimize profitability, and enhance customer satisfaction.

1. Introduction & Business Context

In modern e-commerce and global trade, timely delivery is a critical driver of customer retention and operational profitability. Late deliveries not only erode customer trust but also incur direct financial losses due to operational overheads, rework, and compromised profit margins.

SUPPLY CHAIN OPTIMIZATION

The primary objectives of this project were twofold:

End-to-End Analytics & Predictive Modeling for Delivery Delays

- Diagnostic Analytics:** Quantify the financial impact of delayed orders and identify the root causes (temporal, regional, and categorical) driving these bottlenecks.
- Predictive Analytics:** Develop a robust machine learning classification engine capable of predicting late delivery risk before an order is processed, allowing for preemptive intervention.

2. Data Preprocessing & Methodology

2.1 Data Cleaning & Dimensionality Reduction

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The raw dataset contained extensive transactional data, which required rigorous cleaning to establish a reliable foundation for analysis. Irrelevant, uninformative, redundant, or Personally Identifiable Information (PII) columns were removed. Attributes such as **Customer Email**, **Customer Password**, and **Order Item Cardprod Id** were dropped to reduce noise and ensure data privacy.

Furthermore, records associated with "Shipping canceled" statuses were filtered out, as they represent anomalous supply chain paths that skew standard delivery delay calculations.

2.2 Feature Engineering

To quantify delays and profitability, several critical features were engineered from the raw timestamps and financial data:

- **Order Processing Time:** Calculated as the delta (in days) between the actual shipping date and the order date.
- **Delay Calculation:** Computed as the difference between the `Order Processing Time` and the `Days for shipment (scheduled)`.
- **Is_Delayed Flag:** A boolean indicator created for any order where the Delay was strictly greater than zero.
- **Profitability Matrix:** A categorized feature (`Profitability_Flag`) classifying each order into "Profitable", "Loss", or "Breakeven" using vectorized logic (`np.select`).
- **Temporal Extraction:** Extracted ``order_month``, ``order_day``, and ``order_hour`` to enable granular time-series analysis of supply chain friction.

3. Exploratory Data Analysis & Diagnostic Insights

3.1 Key Performance Indicators (KPIs) & Financial Impact

A specialized formatting algorithm was implemented to present high-level KPIs dynamically. The analysis isolated delayed subsets to reveal the true cost of operational inefficiency. Measuring the `90th Percentile Delay` proved highly effective in establishing realistic "worst-case" scenarios rather than relying on easily skewed averages. By isolating the total loss strictly attributed to delayed orders, the business impact was successfully quantified in monetary terms.

3.2 Temporal & Regional Bottlenecks

A multi-faceted visualization strategy was employed to uncover hidden patterns in delivery performance:

Factor Analysis Highlights:

- **Temporal Frictions:** By aggregating delay percentages across months, days of the week, and hours of the day, specific operational "rush" periods correlated strongly with elevated failure rates in on-time delivery.
- **Regional Drivers:** A programmatic grouping function (`top_drivers_for_region`) was built to dissect delays within specific geographic constraints (e.g., Central Africa). The analysis proved that factors such as *Shipping Mode* and *Department Name* had highly variable success rates depending on the region.

4. Predictive Modeling Framework

Following the descriptive analysis, the project transitioned into building a machine learning pipeline to proactively flag high-risk orders.

4.1 Feature Selection & Categorical Encoding

The feature matrix (`X`) was constructed using both numerical schedules and high-cardinality categorical data (e.g., Category Name, Customer Segment, Order Region). To handle the categorical variables without triggering the curse of dimensionality (which occurs with standard One-Hot Encoding), **Frequency Encoding** was utilized. This technique mapped categories to their normalized frequencies, creating a dense, mathematically sound feature space.

4.2 Handling Class Imbalance with SMOTE

Supply chain datasets inherently suffer from class imbalance; on-time deliveries drastically outnumber late deliveries. Training a model on this skewed distribution biases the algorithm toward predicting the majority class ("On-Time"), resulting in an artificially high accuracy but a disastrously low recall for the actual delays.

To rectify this, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied to the training set. By generating synthetic minority class samples, the model was forced to rigorously learn the boundary conditions of a "Late Delivery," ensuring equitable predictive power.

4.3 Random Forest Classifier Evaluation

A **Random Forest Classifier** was selected for its robustness against non-linear relationships and its inherent resistance to overfitting. The model was trained on the balanced dataset and evaluated on an isolated test set.

Evaluation Metric	Strategic Importance
Accuracy	Provides a baseline overview of total correct predictions across both classes.
Precision	Measures the reliability of a "Late" prediction (minimizing false alarms).
Recall	The most critical metric for this problem—measures the percentage of actual late deliveries the model successfully caught.

5. Conclusion & Strategic Recommendations

This end-to-end analysis successfully bridges the gap between raw supply chain data and actionable predictive intelligence. The transition from identifying historical losses to building a predictive Random Forest Engine represents a significant leap in operational maturity.

Actionable Next Steps:

- **Integrate the Risk Engine:** Deploy the Random Forest model into the order processing pipeline to dynamically flag high-risk orders at checkout.
- **Dynamic Routing:** For orders flagged with a high ``Late_delivery_risk``, automatically route them to faster, more reliable *Shipping Modes*.
- **Targeted Interventions:** Utilize the regional driver insights to renegotiate logistics contracts or increase warehouse staffing in specific regions during historically high-delay months.